Readme

July 5, 2026

1 Grocery Inventory & Sales Performance Dashboard

An end-to-end **Microsoft Excel Data Analytics Project** that transforms raw grocery inventory and sales data into an interactive business dashboard using Excel.

This project demonstrates the complete **Data Analytics Lifecycle**, including data cleaning, data transformation, KPI development, Pivot Tables, Pivot Charts, interactive dashboard design, and business insight generation.

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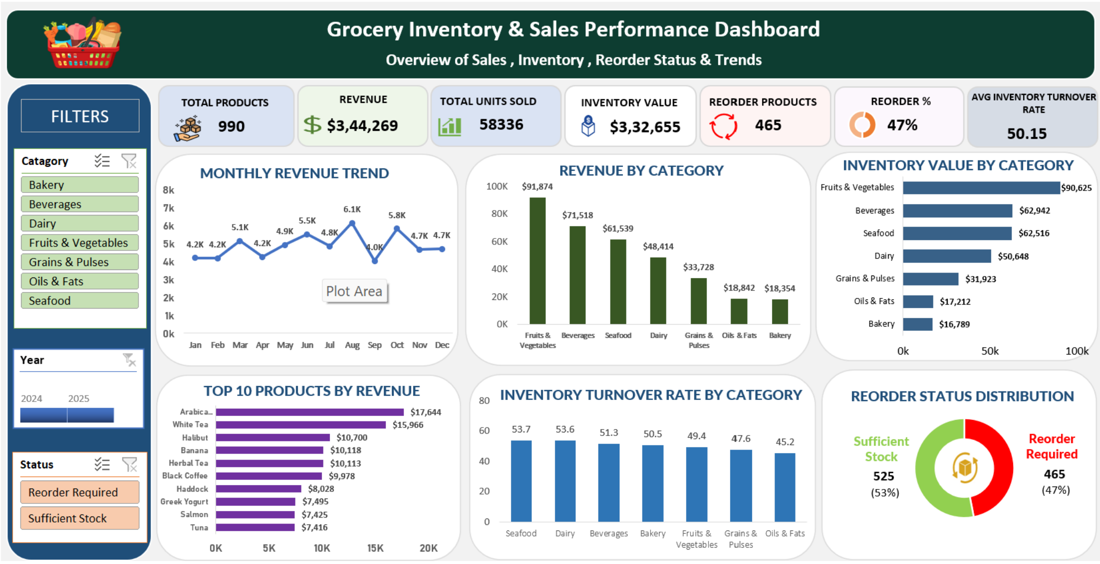
2 Dashboard Preview

Dashboard Screenshot



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3 Project Overview

Managing grocery inventory efficiently is essential for maintaining product availability, reducing inventory costs, and maximizing sales performance.

This dashboard provides an interactive reporting solution that enables business users to monitor inventory health, analyze revenue, identify products requiring replenishment, and evaluate category-level performance.

The project demonstrates practical Microsoft Excel skills commonly used by Data Analysts in real business environments.

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4 Project Objectives

The primary objectives of this project are:

- Analyze overall sales performance.
- Monitor inventory value.
- Track monthly revenue trends.
- Identify products requiring replenishment.
- Compare revenue across categories.
- Identify top-performing products.
- Measure inventory turnover efficiency.
- Develop an interactive business dashboard.

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5 Business Problem

The grocery business maintained inventory and sales records in Excel but lacked a centralized reporting solution.

Business users faced several challenges:

- Manual report preparation
- Difficulty monitoring inventory
- Limited visibility into monthly revenue
- Difficulty identifying reorder products
- No centralized KPI reporting
- Time-consuming business analysis

The objective was to transform raw operational data into an interactive dashboard for faster business decision-making.

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6 Dataset Information

Item	Details
Domain	Grocery Retail
Tool	Microsoft Excel
Total Products	990
Product Categories	7

Item	Details
Analysis Period	2024 – 2025

6.0.1 Main Dataset Columns

- Product_ID
- Product_Name
- Category
- Supplier_ID
- Supplier_Name
- Stock_Quantity
- Reorder_Level
- Reorder_Quantity
- Unit_Price
- Sales_Volume
- Inventory_Turnover_Rate
- Date_Received
- Last_Order_Date
- Expiration_Date

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7 Data Cleaning Process

The raw dataset contained several inconsistencies that required cleaning before analysis.

7.0.1 Data Cleaning Activities

- Standardized mixed date formats.
- Converted Unit Price from text to numeric values.
- Created Revenue column.
- Created Inventory Value column.
- Created Reorder Status column.
- Created Month_Year helper column.
- Validated data types.
- Prepared dataset for Pivot Tables.

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8 Calculated Columns

The following calculated columns were created:

Column	Formula
Revenue	Sales Volume × Unit Price
Inventory Value	Stock Quantity × Unit Price
Reorder Status	IF(Stock < Reorder Level)

Column	Formula
Month_Year	DATE(YEAR(Date),MONTH(Date),1)

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9 Dashboard Features

9.1 KPI Cards

The dashboard includes seven KPI cards:

- Total Products
- Total Revenue
- Total Units Sold
- Inventory Value
- Reorder Products
- Reorder Percentage
- Average Inventory Turnover Rate

9.2 Dashboard Visualizations

The dashboard contains the following visualizations:

9.2.1 Monthly Revenue Trend

Displays monthly revenue performance and helps identify seasonal sales patterns.

9.2.2 Revenue by Category

Compares revenue generated across grocery product categories.

9.2.3 Inventory Value by Category

Displays the monetary value of inventory for each category.

9.2.4 Top 10 Products by Revenue

Ranks products based on revenue generated.

9.2.5 Inventory Turnover Rate by Category

Compares inventory movement efficiency across categories.

9.2.6 Reorder Status Distribution

Shows the percentage of products requiring replenishment versus sufficient stock.

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10 Interactive Filters

The dashboard supports interactive filtering using:

- Category Slicer
- Year Timeline
- Reorder Status Slicer

These filters dynamically update all KPI cards and charts.

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11 Key Performance Indicators (KPIs)

KPI	Description
Total Products	Total number of products
Total Revenue	Total sales revenue
Total Units Sold	Total quantity sold
Inventory Value	Current inventory investment
Reorder Products	Products below reorder level
Reorder Percentage	Percentage of products requiring replenishment
Average Inventory Turnover Rate	Inventory movement efficiency

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12 Business Questions Answered

This dashboard helps answer the following business questions:

- What is the total revenue generated?
- Which category generates the highest revenue?
- Which products generate the highest revenue?
- How does revenue change month by month?
- What is the current inventory value?
- Which category has the highest inventory investment?
- Which products require replenishment?
- What percentage of products require reorder?

- Which category has the highest inventory turnover?

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13 Business Insights

The dashboard provides insights into:

- Overall business performance
- Category-wise revenue
- Monthly revenue trends
- Inventory investment
- Inventory turnover
- Product performance
- Reorder requirements

These insights help management make informed inventory and sales decisions.

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14 Tools & Technologies

- Microsoft Excel
- Excel Tables
- Pivot Tables
- Pivot Charts
- Excel Formulas
- Slicers
- Timeline Filters
- Conditional Formatting
- Dashboard Design

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15 Skills Demonstrated

15.1 Technical Skills

- Microsoft Excel
- Data Cleaning
- Data Transformation
- Advanced Excel Formulas
- Pivot Tables
- Pivot Charts
- Dashboard Design
- KPI Development
- Data Validation

15.2 Analytical Skills

- Business Analysis
- Inventory Analysis
- Sales Analysis
- Revenue Analysis
- Data Visualization
- Dashboard Development
- Decision Support

15.3 Professional Skills

- Problem Solving
- Analytical Thinking
- Business Communication
- Documentation
- Presentation Skills

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16 Repository Structure

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Grocery-Inventory-Sales-Performance-Dashboard

  Dashboard/
     Grocery_Inventory_Dashboard.xlsx
     Dashboard.png

  Dataset/
     Grocery_Inventory_Dataset.csv
     Cleaned_Dataset.xlsx

  Documentation/
     01_Business_Requirement_Document.md
     02_Domain_Knowledge.md
     03_Data_Cleaning_Process.md
     04_Project_Report.md
     05_Interview_Preparation.md

  Images/
     Dashboard.png

  README.md
```

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17 Future Enhancements

Future improvements may include:

- Profit & Margin Analysis
- Supplier Performance Dashboard
- Expiry Risk Monitoring
- Reorder Forecasting
- SQL Database Integration
- Power BI Dashboard
- Python-based Predictive Analytics

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18 Project Documentation

This repository includes complete project documentation:

- Business Requirement Document
- Domain Knowledge
- Data Cleaning Process
- Project Report
- Interview Preparation Guide

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19 About the Project

This project was developed as part of my Data Analytics portfolio to demonstrate practical Microsoft Excel skills used in business reporting and dashboard development.

It showcases the complete workflow of a Data Analyst—from understanding business requirements and cleaning data to building an interactive dashboard and generating business insights.

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20 Author

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20.0.1 Connect with Me

- LinkedIn: *(www.linkedin.com/in/nikhil-ramagiri-21b2a324a)*
- GitHub: *(https://github.com/RamagiriNikhil/grocery-inventory-sales-performance-dashboard)*

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21 Support

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Your support is greatly appreciated!

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